

HABLA 2.0

Progress Update

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SCOPE

Seven attack pipelines (FishGram, Qwen3-TTS, Chatterbox, OpenVoice, OuteTTS, CosyVoice) plus a novel word-level partial spoof pipeline. 1,567 speakers across 7 Latin American accents. Three pipelines at production scale (~35K samples each).

Two gaps, one corpus

Voice anti-spoofing decides whether a speech signal is genuine (*bonafide*) or machine-generated (*spoofed*). The reference resource for Latin American Spanish — HABLA (Interspeech 2023) — is three years stale, and the field still frames spoofing as all-or-nothing. HABLA 2.0 closes both gaps at once.

GAP A — AN OUTDATED CORPUS

HABLA is 3+ years old

GAN-only attacks (StarGAN-VC, CycleGAN-VC) — deprecated 2018–2020 technology.

No augmentation — models trained on clean studio audio collapse under reverb, codec compression, and noise.

No neural codec / flow-matching TTS — the dominant 2025–2026 synthesis paradigm is entirely absent.

English detectors hit EER >45% on Spanish — cross-lingual transfer fails.

GAP B — THE PARTIAL-SPOOF BLIND SPOT NEW

Every corpus is all-or-nothing

All anti-spoofing datasets — HABLA included — contain **fully-synthetic OR fully-bonafide** audio only.

The real threat is **partial**: splice a few cloned words into genuine speech ("transfer 100"→"transfer 9000"). 90%+ of the signal is real, so global detectors miss it.

No Latin American Spanish partial-spoof corpus exists.

Detectors learn a **shortcut** — the noisy splice boundary, not real synthesis artifacts (Negroni 2024: 6.16% EER, zero training). It does not generalize.

HABLA's regional accents remain valuable. Its attack set, its lack of augmentation, and its all-or-nothing framing do not. HABLA 2.0 modernizes the attacks, hardens robustness, and introduces the first Spanish partial-spoof corpus with adversarial boundary augmentation.

One goal, four pillars

Design and implement **HABLA 2.0**: a modernized, multi-condition voice anti-spoofing corpus and benchmark for Latin American Spanish, built on four pillars — modern full-synthesis attacks, a partial-spoof corpus with boundary jitter, multi-condition acoustic augmentation, and cross-corpus benchmarking.

PILLAR 1

Full-synthesis attacks

6 neural TTS systems (Fish Speech, Qwen3, OmniVoice, Chatterbox, OpenVoice, OuteTTS) replace the legacy GAN attacks.

MOSTLY DONE

PILLAR 2

NEW

Partial spoof

Word-level splicing of 1–3 cloned words (W1/W2/W3) plus boundary-jitter adversarial augmentation. First in Spanish.

VALIDATED

PILLAR 3

Acoustic augmentation

Multi-condition degradation: 60% RIR+Noise / 30% Codec / 10% RawBoost, for real-world robustness.

PENDING

PILLAR 4

Cross-corpus benchmarking

SSL detectors (wav2vec 2.0 / WavLM) over full + partial spoof; evaluate vs ASVspoof, HISPASpoof, PartialSpoof.

PENDING

- 0E1** Speaker-independent partitions (HABLA + Common Voice), no overlap across train/val/test.

0E2 Acoustic augmentation pipeline (60/30/10 RIR-Noise / Codec / RawBoost).

PENDING

0E3 Full-synthesis attacks with 6 modern neural TTS systems.

0E4 Partial-spoof corpus: word-level splice (clone → align → valley-select → splice), ECAPA gate.

NEW

0E5 Boundary-jitter adversarial augmentation: homogenize artifacts across all boundaries to break the detector shortcut.

NEW

0E6 Train and evaluate detectors (SSL, AM-Softmax) on full + partial spoof vs LCNN+LFCC baseline.

PENDING

0E7 Cross-corpus benchmarking vs ASVspoof 2019 LA, HISPASpoof, PartialSpoof.

PENDING

Where the four pillars stand

Pillars 1 and 2 are implemented and validated. Pillars 3 and 4 — acoustic augmentation and the detector/benchmark suite — are the remaining thesis work.

PILLAR 1 — FULL-SYNTHESIS ATTACKS

MOSTLY DONE

FishGram (Fish Speech 4B)	PROD 95.2%
Qwen3-TTS	PROD 87.9%
OpenVoice	PROD 83.4%
OmniVoice (diffusion LM)	VALIDATED
Chatterbox / OuteTTS	RUNNING

PILLAR 3 — ACOUSTIC AUGMENTATION

PENDING

Multi-condition pipeline (60% RIR+Noise / 30% Codec / 10% RawBoost) designed and grounded in the ASVspoof 5 winners' distribution, but **not yet applied to the corpus**. Augmenters scaffolded; integration into the production sweep is the next step.

PILLAR 2 — PARTIAL SPOOF

VALIDATED

7-step pipeline complete — transcribe → clone → align → valley-select → splice → validate → format, 6 attack backends.

Splice engine validated (2026-05-24): natural-duration splicing, energy-refined boundaries, asymmetric valley snap. Audit auditiva passed.

Boundary jitter implemented (truncate / overlap / bleed, p=0.5).

Manifest dispatch + parallel orchestrator, 6 attacks × 2 partitions.

PILLAR 4 — CROSS-CORPUS BENCHMARKING

PENDING

SSL detector training (wav2vec 2.0 / WavLM, AM-Softmax) over full + partial spoof, then evaluation vs ASVspoof 2019 LA, HISPASpoof, and PartialSpoof. **Depends on the full production sweep finishing first.**

6

NEURAL TTS ATTACKS
INTEGRATED

1,567

SPEAKERS, 7 LATAM ACCENTS

295

PARTIAL-SPOOF VALIDATION
SAMPLES

2/4

PILLARS COMPLETE

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next →

Nine pipelines

All share the same architecture — Facade orchestrator, Strategy steps, Pydantic schemas, loguru logging. Seven distinct TTS codec architectures ensure the detector cannot overfit to a single synthesis fingerprint. Plus a novel word-level partial spoof pipeline.

PIPELINE 01 Mozilla Speaker Selection select_mozilla_speakers/ Expands the bonafide dataset with CV v24.0 speakers — adds Mexico and Spain for the first time. COMPLETE	PIPELINE 02 FishGram Attack fishgram_attack/ Zero-shot cloning via Fish Speech 4B (VQGAN). HTTP API on ml-server03. PRODUCTION	PIPELINE 03 Qwen Attack qwen_attack/ Qwen3-TTS 1.7B (Dual-Track codec). Local inference, STT-enhanced reference. PRODUCTION	PIPELINE 04 Chatterbox Attack chatterbox_attack/ Chatterbox 500M (flow-matching). Trailing-noise post-processing. RUNNING	PIPELINE 05 OpenVoice Attack openvoice_attack/ MeloTTS + ToneColorConverter. Two-stage tone-color transfer. PRODUCTION
PIPELINE 06 OuteTTS Attack outetts_attack/ OuteTTS 1.0 0.6B (Qwen3 + DAC codec). LLM-based, 14 languages incl. Spanish. VALIDATED	PIPELINE 07 OmniVoice Attack omnivoice_attack/ k2-fsa OmniVoice (diffusion LM, 646 languages, 27,559 h Spanish). Custom prefix-bleed detection + retry loop. VALIDATED	PIPELINE 08 CosyVoice cosyvoice_attack/ CosyVoice2-0.5B (Conditional Flow Matching). No Spanish support — generates Chinese. DROPPED	PIPELINE 09 — NOVEL CONTRIBUTION Partial Spoof partial_spoof/ Word-level splicing: replaces 1-3 words in bonafide audio with TTS-cloned versions. 7-step pipeline with 6 attack backends. First partial spoof for Latin American Spanish. TESTING	

5

STEPS PER PIPELINE

7

TTS ARCHITECTURES

7

ACCENT REGIONS

1,567

SPEAKERS IN HABLA V2

Fish Speech 4B (OpenAudio S1-mini)

HTTP API-based TTS with VQGAN vocoder and zero-shot voice cloning. The largest model in our attack suite at 4 billion parameters, requiring only 15 seconds of reference audio to clone a speaker.

4B

MODEL SIZE

VQGAN

ARCHITECTURE

HTTP API

INFERENCE

15s

REFERENCE AUDIO

First pipeline to reach production. FishGram is present in both LRLSpoof and HispaSpoof, enabling direct cross-dataset comparison against our HABLA 2.0 results. Its high pass rate (95.2%) and NISQA (4.571) make it the most reliable generator in the suite.

CosyVoice 2: no Spanish support

CosyVoice2-0.5B (Conditional Flow Matching) was evaluated but generates Chinese phonemes when given Spanish input. Supports only zh/en/ja/yue/ko. Near real-time speed but fundamentally unable to produce Spanish speech. Dropped from the attack suite.

PROPERTY	VALUE
Model	CosyVoice2-0.5B
Architecture	Conditional Flow Matching
Speed	RTF 0.77 - 0.99
Languages	zh / en / ja / yue / ko
Spanish support	NO
Status	DROPPED

Valid negative finding. Documents that not all multilingual TTS systems handle Spanish despite marketing claims. CosyVoice 2 defaults to Chinese phonemization for unsupported languages, producing unintelligible output for Spanish input text.

OuteTTS 1.0 0.6B (Qwen3 + DAC)

LLM-based TTS using a Qwen3 0.6B backbone with DAC audio codec. Supports 14 languages including Spanish, trained on MLS and Common Voice. Autoregressive generation yields high audio quality at the cost of speed. Apache 2.0 license.

0.6B MODEL SIZE	LLM + DAC codec ARCHITECTURE	~6x RTF SPEED (SLOW)
3s REFERENCE AUDIO	14 LANGUAGES (INCL. SPANISH)	Apache 2.0 LICENSE

Novel architecture. Absent from both LRLSpooF and HispaSpooF. The LLM backbone (Qwen3) represents a fundamentally different approach to speech synthesis compared to flow-matching or VQGAN methods, adding architectural diversity to our attack suite.

OmniVoice: 646-language diffusion TTS with reference-bleed handling

Diffusion language model from k2-fsa supporting 646 languages, with 27,559 hours of Spanish training data — the largest Spanish coverage in any open zero-shot TTS we evaluated. Required custom post-processing because diffusion sampling occasionally bleeds the reference voice into the leading audio frames in a way that ASR-based detection cannot see.

Diffusion LM

ARCHITECTURE

646

LANGUAGES

27,559 h

SPANISH TRAINING DATA

10s

REFERENCE AUDIO

RTF ~0.025

SPEED (40× REAL-TIME)

24 kHz native

SAMPLE RATE (RESAMPLED TO 16 KHZ)

REFERENCE VOICE BLEED: AN ARTIFACT ASR CANNOT SEE

Diffusion sampling occasionally emits a fragment of the reference voice (200–600 ms) before the prompt content begins. The fragment is sub-syllabic, so Parakeet TDT does not transcribe it — WER stays at 0.0 even though the audio is clearly contaminated. Word-alignment-based prefix detectors (used for Qwen3-TTS hallucinated words) are blind to this failure mode.

EMPIRICAL SIGNAL (VALIDATION SET, 6 SAMPLES)

Artifact samples:

T_first ≈ 0.56–0.64 s

pre_RMS ≈ -22 to -25 dB

(only 2–5 dB below speech level)

Clean samples:

T_first = 0.0 – 0.08 s

pre_RMS = -120 dB

(silence floor)

DISCRIMINATOR MARGIN

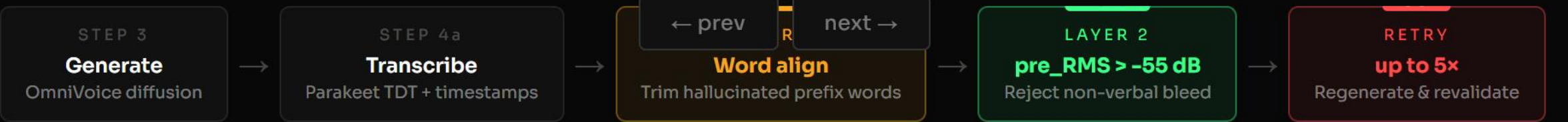
Threshold of **-55 dBFS** sits

+30 dB above the artifact band

+65 dB below the silence band

```
pre_rms_db = 20 * log10(rms[0:T_first])
if pre_rms_db > -55: reject
```

TWO-LAYER DETECTION + RETRY LOOP



Root cause identified and fixed. The bleed traced to `audio_concatenation.py` slicing the last bonafide file mid-word to hit a 10 s target. OmniVoice's diffusion conditioning attempted to "complete" the cut-off pattern at the start of generation. Fix: stop at the last fitting file, snap to silence in the edge case, always append 200 ms trailing silence so the reference ends cleanly. Post-fix validation: **6/6 passed first attempt**, zero non-verbal-prefix rejections, zero retries needed. `anf_00295` -- previously deterministic-fail across 6 attempts -- now passes cleanly. Detector + retry kept as defense in depth.

FishGram at scale

Full production run: 1,567 speakers x ~23 samples each. Fish Speech (OpenAudio S1-mini, VQGAN vocoder) running as persistent HTTP API on ml-server03 GPU

3. Quality validated via Parakeet STT + NISQA + ECAPA-TDNN.

35,927

Samples generated

1,567 speakers

34,197

Passed validation

95.2% pass rate

1,730

Rejected

4.8% rejection rate

36.1h

Total audio

avg 3.80s per sample

QUALITY METRICS (PASSED SAMPLES)

METRIC	MEAN	MEDIAN	STD
WER	2.17%	0.00%	4.26%
CER	0.70%	0.00%	1.62%
NISQA MOS	4.571	4.890	0.764
Speaker similarity	0.602	0.618	0.112

Scale vs validation shift. At production scale, mean similarity dropped from 0.567 (validation, 6 samples) to 0.602 (production, 34K samples) — actually *improved* with more speakers. NISQA dropped from 4.978 to 4.571 as more diverse and challenging texts were generated, but median remains high at 4.890.

REJECTION BREAKDOWN

WER only

1,341

77.5% of rejections — pronunciation errors on proper nouns and rare words

WER + CER

378

21.8% — both word and character errors exceeded thresholds simultaneously

Near-silent output

11

0.6% — model produced near-silent or garbled audio (RMS below threshold)

Quality distributions

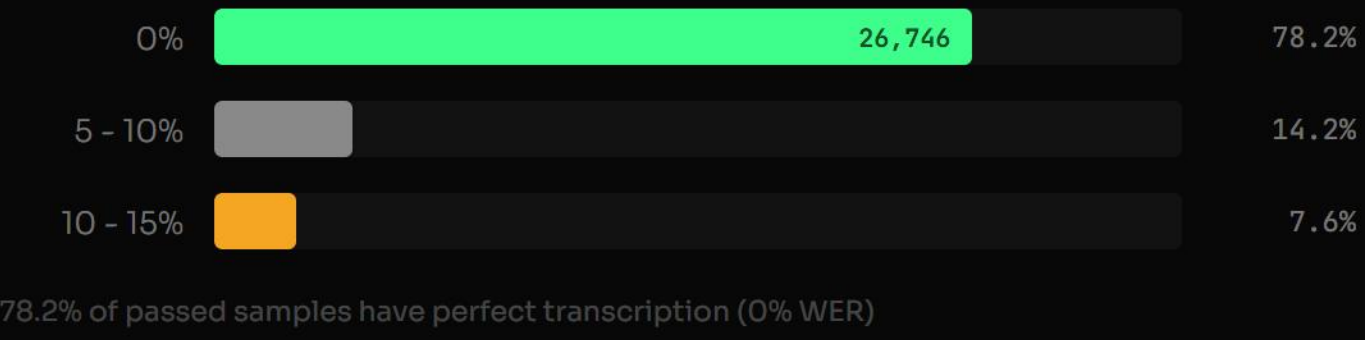
Distribution of NISQA MOS, speaker similarity, WER, and per-accent pass rates across all 34,197 passed samples.

NISQA MOS DISTRIBUTION



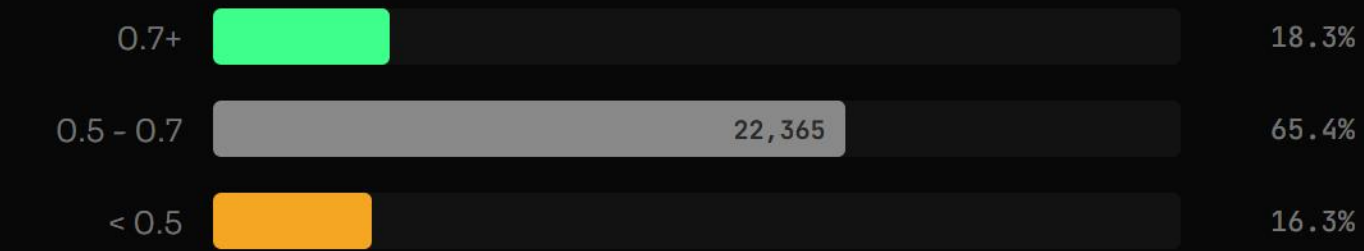
Median: 4.890 — three quarters of samples achieve near-perfect perceptual quality

WER DISTRIBUTION (PASSED)



78.2% of passed samples have perfect transcription (0% WER)

SPEAKER SIMILARITY DISTRIBUTION



Mean: 0.602 — bulk of clones land in the moderate fidelity zone (0.5-0.7)

PASS RATE BY ACCENT



Accent-uniform quality — Spain (ES) slightly lower, likely due to phonetic diversity in text prompts

Production insight. Pass rates are remarkably uniform across all 7 accents (93.2% - 96.0%), confirming that Fish Speech's VQGAN architecture handles Latin American Spanish accent diversity well. The slight ES dip may reflect Castilian-specific phonemes in the Common Voice text corpus that challenge the model.

Qwen3-TTS at scale

Full production run: 1,567 speakers. Qwen3-TTS 1.7B local model with voice cloning via x-vector prompts. Reference transcripts from faster-whisper. Quality validated via Parakeet STT + NISQA + ECAPA-TDNN.

35,927

Samples generated

1,567 speakers

31,568

Passed validation

87.9% pass rate

4,359

Rejected

12.1% rejection rate

40.1h

Total audio

avg 4.57s per sample

QUALITY METRICS (PASSED SAMPLES)

METRIC	MEAN	MEDIAN	STD
WER	1.46%	0.00%	3.58%
CER	0.59%	0.00%	1.73%
NISQA MOS	4.371	4.628	0.676
Speaker similarity	0.720	0.737	0.095

Highest speaker similarity. Qwen3-TTS achieves the best speaker similarity across all pipelines (0.720 mean), outperforming FishGram (0.602). The x-vector voice cloning mechanism captures speaker characteristics more faithfully, making Qwen attacks the most dangerous for embedding-based detectors.

REJECTION BREAKDOWN

WER + CER

65.1% of rejections — both thresholds exceeded simultaneously

2,839

WER only

20.3% — word-level pronunciation errors on complex vocabulary

887

CER only

13.0% — character-level substitutions below WER threshold

568

Artifacts (silent/truncated)

1.5% — near-silent or truncated output detected by artifact checks

65

OpenVoice at scale

Full production run: 1,567 speakers. OpenVoice V2 (MeloTTS base + ToneColorConverter at tau=0.3). Fastest pipeline at RTF 0.07-0.10 (10-14x real-time). Quality validated via Parakeet STT + NISQA + ECAPA-TDNN.

35,544

Samples generated

1,567 speakers

29,626

Passed validation

83.4% pass rate

5,918

Rejected

16.6% rejection rate

37.2h

Total audio

avg 4.52s per sample

QUALITY METRICS (PASSED SAMPLES)

METRIC	MEAN	MEDIAN	STD
WER	1.50%	0.00%	3.67%
CER	0.45%	0.00%	1.28%
NISQA MOS	4.409	4.556	0.542
Speaker similarity	0.394	0.397	0.107

Lowest speaker similarity. OpenVoice's tone-color conversion at tau=0.3 only partially transfers speaker characteristics (0.394 mean), less than half of Qwen's 0.720. The MeloTTS base voice "bleeds through." However, it achieves the lowest CER (0.45%) and highest consistency (lowest std on NISQA). For anti-spoofing, OpenVoice attacks are the easiest for embedding-based detectors to catch.

REJECTION BREAKDOWN

WER only

48.5% of rejections — MeloTTS sentence splitting causes word-level errors

2,873

Other (duration/format)

32.4% — duration anomalies from MeloTTS sentence chunking artifacts

1,919

WER + CER

17.1% — both thresholds exceeded simultaneously

1,009

CER only

2.0% — minor character-level errors

117

Chatterbox (Resemble.ai)

IN PROGRESS

Full production run: 1,567 speakers on GPU 2. Chatterbox uses a GPT-style autoregressive decoder with classifier-free guidance (cfg=0.5) and a watermark removal step. Significantly slower than other pipelines due to high RTF.

21,192

Samples generated
of 35,927 target

59%

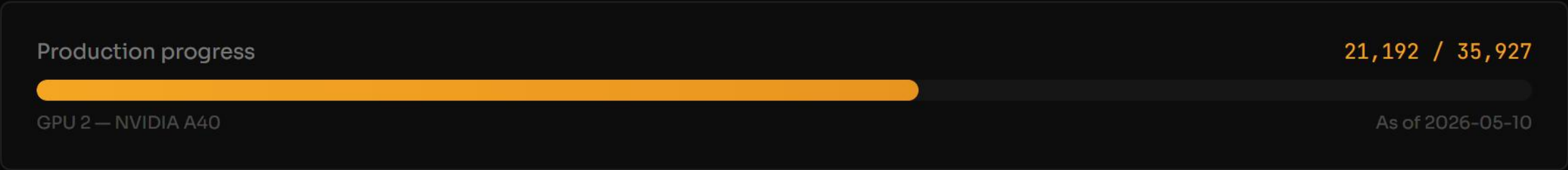
Progress
Started April 12

31-45x

RTF range
Slowest pipeline

~early June

ETA
~14,700 remaining



PIPELINE CHARACTERISTICS

Architecture

GPT-style autoregressive TTS with CFG (classifier-free guidance). Generates EnCodec tokens then decodes via Vocos vocoder.

Voice cloning

Zero-shot via 15s reference audio + speaker embedding. No fine-tuning required. Watermark removal enabled.

OBSERVATIONS SO FAR

RTF 31-45x is prohibitive.

A 5-second utterance takes 150-225 seconds to generate. At this rate, the full 35,927 samples require ~31 days of continuous GPU time. Chatterbox is the slowest of all 5 pipelines by a wide margin (FishGram: 2-3x, Qwen: 3-5x, OpenVoice: 0.07-0.10x).

Quality expectations:

Resemble.ai reports high naturalness in English. Spanish quality is unvalidated until the run completes. Early samples show reasonable intelligibility but speaker similarity is TBD.

OuteTTS at scale

Full production run: 1,567 speakers. OuteTTS 1.0 0.6B (Llama 3.1 backbone + WavTokenizer codes), zero-shot voice cloning via 15s speaker prompt tokens. Quality validated via Parakeet STT + NISQA + ECAPA-TDNN. Moderate RTF ~5.6x.

35,461

Samples generated

1,567 speakers

25,642

Passed validation

72.3% pass rate

9,819

Rejected

27.7% rejection rate

30.9h

Total audio

avg 4.33s per sample

QUALITY METRICS (PASSED SAMPLES)

METRIC	MEAN	MEDIAN	STD
WER	2.36%	0.00%	4.36%
CER	0.85%	0.00%	1.84%
NISQA MOS	4.453	4.625	0.534
Speaker similarity	0.462	0.471	0.115

Lower speaker similarity than other pipelines. OuteTTS's 0.462 mean ECAPA SIM is well below Qwen (0.720) and FishGram (0.602), comparable to OpenVoice (0.394). The WavTokenizer-based reproduction preserves linguistic content well (NISQA 4.45 -- second only to FishGram) but speaker timbre transfer is weaker.

REJECTION BREAKDOWN

WER + CER

5,968

60.8% of rejections — both thresholds exceeded simultaneously

WER only

3,375

34.4% — word-level pronunciation errors on complex vocabulary

CER only

262

2.7% — character-level substitutions below WER threshold

Near-silent / artifacts

214

2.2% — near-silent output (213) + duration anomaly (1)

OmniVoice at scale

Full production run: 1,567 speakers. OmniVoice (k2-fsa) diffusion language model TTS, 646 languages, 27,559 h Spanish training data. Zero-shot voice cloning via 10s reference. Quality validated via Parakeet STT + NISQA + ECAPA-TDNN, with a non-verbal prefix detector and 5-round retry loop guarding against reference-bleed artifacts.

35,927

Samples generated

1,567 speakers

33,743

Passed validation

93.9% pass rate (best of suite)

2,184

Rejected

6.1% rejection rate

39.3h

Total audio

avg 4.19s per sample

QUALITY METRICS (PASSED SAMPLES)

METRIC	MEAN	MEDIAN	STD
WER	1.06%	0.00%	3.16%
CER	0.30%	0.00%	1.05%
NISQA MOS	4.284	4.657	0.888
Speaker similarity	0.686	0.705	0.109

Lowest WER of the suite (1.06%) and highest pass rate (93.9%). OmniVoice's diffusion conditioning produces highly intelligible Spanish despite being a non-autoregressive architecture. Speaker similarity 0.686 sits between Qwen (0.720) and FishGram (0.602) -- a strong all-around attack profile.

REJECTION BREAKDOWN

Near-silent / artifacts

45.3% — near-silent (979) + duration anomaly (10)

989

WER only

31.1% — word-level pronunciation errors

679

Non-verbal prefix artifact

16.8% — reference-bleed caught by custom RMS detector (defense in depth)

367

WER + CER

6.4% — both thresholds exceeded simultaneously

139

CER only

0.5% — character-level substitutions below WER threshold

10

Cross-pipeline comparison

Production metrics from ~35K samples each (1,567 speakers). FishGram, Qwen3-TTS, OpenVoice, OuteTTS, and OmniVoice complete. Chatterbox in progress (21,192 / 35,927, 59%). CosyVoice dropped (no Spanish support).

Metric	FishGram	Qwen3TTS	OpenVoice	Chatterbox	OuteTTS	OmniVoice
Architecture	VQGAN vocoder	Dual-Track codec	VITS tone-color	Flow-matching	LLM + DAC codec	Diffusion LM
Model size	4B	1.7B	~200M	500M	0.6B	~1.5B
Samples generated	35,927	35,927	35,544	21,192*	35,461	35,927
Passed validation	34,197	31,568	29,626	—	25,642	33,743
Pass rate	95.2%	87.9%	83.4%	—	72.3%	93.9%
Avg WER	2.17%	1.46%	1.50%	—	2.36%	1.06%
Avg CER	0.70%	0.59%	0.45%	—	0.85%	0.30%
Avg NISQA MOS	4.571	4.371	4.409	—	4.453	4.284
Avg speaker similarity	0.602	0.720	0.394	—	0.462	0.686
Total audio (passed)	36.1h	40.1h	37.2h	—	30.9h	39.3h

* Chatterbox still running as of 2026-05-10 (RTF 31-45x); ETA early June.

Production-scale findings. OmniVoice achieves the lowest WER (1.06%) and CER (0.30%) of the suite at 93.9% pass rate — competitive with FishGram on quality, with stronger speaker fidelity than FishGram (0.686 vs 0.602). Qwen3-TTS remains the most dangerous attack by speaker similarity (0.720). FishGram still leads on pass rate (95.2%) and NISQA (4.571). OpenVoice and OuteTTS have the weakest voice cloning (SIM 0.394 and 0.462) — easiest for embedding-based detectors. The 6 architectures span 200M to 4B parameters and 5 distinct synthesis paradigms (VQGAN, LM+codec, VITS-style, flow-matching, LM+wavtokenizer, diffusion LM) — the diversity ensures detectors cannot overfit to a single synthesis paradigm.

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Rejected samples across pipelines

Examples of rejected samples from FishGram, Chatterbox, Qwen3-TTS, and OpenVoice. Each pipeline struggles with different text — proper nouns, foreign words, and complex vocabulary cause the highest WER.

BONAFIDE REFERENCE — arf_00295

arf_00295

Argentina · Female · 15s reference clip

BONAFIDE

▶ 0:00 / 0:10

🔊

⋮

REJECTED — "EURIDICE" PROPER NOUN

"Euridice huyo, pero fue capturada y hecha prisionera."

REJECTED

CHATTERBOX · arf_00295

▶ 0:00 / 0:03

🔊

⋮

ASR heard:
"Eury se huyo pero fue capturada y hecha prisionera."

WER

25.0%

THRESHOLD

15.0%

REJECTED

FISHGRAM · arf_00295

▶ 0:00 / 0:03

🔊

⋮

ASR heard:
"Yobidi se huyo pero fue capturada y hecha prisionera."

WER

25.0%

THRESHOLD

15.0%

REJECTED — PRODUCTION RUN EXAMPLES

REJECTED

QWEN3TTS · arf_00295

▶ 0:00 / 0:05

🔊

⋮

ASR heard:
"In la cocina, Onsen Riori, la sopa miso, la sopa principal."

WER

36.4%

CER

10.3%

REJECTED

OPENVOICE · arf_00295

▶ 0:00 / 0:03

🔊

⋮

ASR heard:
"Tuvo de defensa y su ultimo equipo fue el Motawa."

WER

20.0%

CER

8.2%

← prev

Common pattern. Rejections cluster around proper nouns, foreign words, and rare vocabulary. Chatterbox/FishGram split "Euridice" differently. Qwen mangles Japanese cuisine terms ("Onsen Riori"). OpenVoice mishandles sports team names ("Motawa"). The WER threshold (15%) catches these consistently across all pipelines.

next →

PRODUCTION PASS RATES

95.2%

87.9%

83.4%

FishGram

Qwen

OpenVoice

PRIMARY REJECTION CAUSE

WER-based (all pipelines)

Proper nouns and out-of-vocabulary terms drive most rejections

Bonafide vs OuteTTS

OuteTTS 1.0 0.6B uses a Qwen3 backbone with DAC audio codec. Trained on 14 languages including Spanish (MLS + Common Voice). High audio quality but moderate speaker similarity due to short 3s reference constraint.

ORIGINAL · BONAFIDE

arf_00295

Argentina · Female · 3s reference

▶ 0:00 / 0:03

SYNTHETIC · SPOOF **OUTETTS**

arf_00295 — clone

Qwen3 0.6B · DAC codec · temp=0.4

▶ 0:00 / 0:05

SPEAKER SIMILARITY

0.534

ECAPA-TDNN cosine · moderate

NISQA MOS

4.46

High perceived quality

WER

0.00%

Perfect intelligibility

OuteTTS trade-off: Excellent audio quality (NISQA 4.46–4.76 across validation) and good intelligibility (0% WER on this sample), but moderate speaker similarity (avg 0.47). The 3-second reference window — constrained by the 8,192-token context — limits how much speaker identity can be captured. Generation is slow (~6x RTF) due to autoregressive LLM decoding. Still, it represents a novel LLM+codec architecture absent from LRLSpooof and HispaSpooof.

CosyVoice: no Spanish support

CosyVoice 2 (0.5B, Conditional Flow Matching) was evaluated as a potential attack system. Despite near real-time generation speed (RTF 0.77-0.99), it **does not support Spanish**. The model generates Chinese-sounding audio when given Spanish text input. Dropped from the attack suite.

ORIGINAL · BONAFIDE

arf_00295

Argentina · Female · Spanish

▶

0:00 / 0:15

🔊

⋮

FAILED · CHINESE OUTPUT

COSYVOICE

arf_00295 — attempted clone

CosyVoice2-0.5B · generates Chinese, not Spanish

▶

0:00 / 0:13

🔊

⋮

Why it failed: CosyVoice2-0.5B only supports 5 languages: Chinese (zh), English (en), Japanese (ja), Cantonese (yue), and Korean (ko). When given Spanish text, the model maps phonemes to its Chinese acoustic space, producing Chinese-sounding speech. This is a fundamental model limitation, not a configuration issue.

COSYVOICE SUMMARY	
Model	CosyVoice2-0.5B
Architecture	Conditional Flow Matching
Languages	zh, en, ja, yue, ko only
Speed (RTF)	0.77-0.99 (near real-time)
Status	DROPPED
Reason	Oriental accent bias, no Spanish

OmniVoice: Spanish voice cloning

OmniVoice was evaluated as an additional attack system for Latin American Spanish. Unlike CosyVoice, it produces fluent Spanish output with strong intelligibility. Sample below: speaker arf_00295 (Argentina Female), single-shot clone from a reference utterance.

ORIGINAL · BONAFIDE

arf_00295
Argentina · Female · Spanish

▶ 0:00 / 0:06

🔊 ⋮

CLONED · OMNIVOICE OMNIVOICE

arf_00295 — synthetic
"Fue profesor de literatura, poeta, periodista de prensa y television."

▶ 0:00 / 0:05

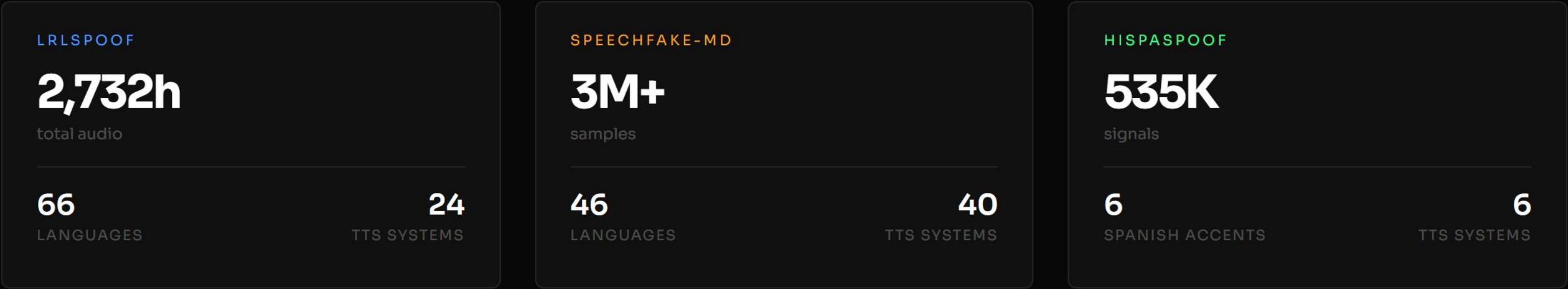
🔊 ⋮

Why it works: OmniVoice handles Spanish phonology natively and preserves the reference speaker's timbre with high fidelity. RTF below 0.35 makes it competitive with Qwen3-TTS on speed while producing cleaner short utterances.

OMNIVOICE VALIDATION (ARF_00295 / SAMPLE 2)	
Duration	5.14s
RTF	0.20 (fast)
WER · CER	0.0% · 0.0%
NISQA MOS	4.99 (high quality)
Speaker similarity	0.714
Status	VIABLE

Datasets investigation

Survey of existing anti-spoofing and synthetic speech datasets. Three key datasets compared against our HABLA 2.0 approach: LRLSpooF (multilingual, 66 languages), SpeechFake-MD (large-scale deepfake), and HispaSpooF (Spanish-specific).



Research gaps. All three datasets lack Latin American accent granularity, partial spoof attacks, and per-sample quality validation — the three core gaps our thesis addresses with HABLA 2.0.

LRLSpoof: Low-Resource Language Spoofing Corpus

2,732 hours of synthetic speech across 66 languages and 24 TTS systems. Submitted to Interspeech 2026 (arXiv:2603.02364). Only synthetic audio — no bonafide samples. Evaluation uses threshold transfer from ASVspoof 2021.

DATASET STRUCTURE

SCALE

2,732h

66

24

Total audio

Languages

TTS systems

SPANISH PORTION

23.14 hours

11 TTS systems used for Spanish · no accent differentiation

TTS OVERLAP WITH OUR WORK

FISH SPEECH

CHATTERBOX

MELOTTTS

OUTETTS

3 of our 4 systems + MeloTTS (used by OpenVoice). They do NOT have Qwen3-TTS.

SPANISH SPOOF REJECTION RATES (SRR)

COUNTERMEASURE	SPANISH SRR
w2v2_300	100.00%
aasist3	99.83%
df_arena_500	99.99%
rescapsguard	99.98%
df_arena_1b	31.64%
res2tcn	26.28%
ssl_aasist	0.61%
nes2net	0.00%

Massive cross-model disparity. Some detectors achieve near-perfect Spanish detection (w2v2_300: 100%), while others completely fail (nes2net: 0%). Detector choice matters enormously — this supports our motivation.

SpeechFake: Multilingual Deepfake Dataset

3+ million deepfake samples across 46 languages from 40 TTS tools. Published at ACL 2025 (arXiv:2507.21463). Spanish is one of 9 primary languages. Bonafide sourced from Mozilla CommonVoice — same as our text pipeline.

DATASET STRUCTURE

SCALE

3M+

Deepfake samples

3,000h

Total audio

40

TTS tools

SPANISH DETECTION (W2V+AASIST)

0.12–0.42%

Spanish EER range

0.15%

English EER (best)

KEY FINDING

"Language content does affect detection performance, even when the generation methods are seen during training."

HABLA 2.0 VS RELATED DATASETS

ASPECT	HABLA 2.0 (OURS)	LRLSPOOF	SPEECHFAKE
Focus	LatAm Spanish	66 languages	46 languages
Accent detail	7 accents	None	None
Bonafide data	Yes (HABLA speakers)	No	CommonVoice
TTS systems	4 (+2 planned)	24	40
Attack type	Voice cloning	General TTS	TTS + VC
Quality metrics	WER/CER/NISQA/SIM	SRR only	EER only
Speakers	1,567	—	—

Our novel contributions. (1) Latin American accent-level granularity (7 accents vs undifferentiated Spanish). (2) Paired bonafide + spoofed data from same speaker pool. (3) Qwen3-TTS — absent from both LRLSpooF and SpeechFake. (4) Per-sample quality validation pipeline (WER, CER, NISQA, speaker similarity).

HISPASpoof: Spanish Speech Forensics

535,687 speech signals across 6 Spanish accents and 6 TTS systems. From Purdue VIPER Lab (Risques et al., arXiv:2509.09155). The first large-scale Spanish-specific anti-spoofing dataset — and the closest related work to our thesis.

CRITICAL FINDING: ENGLISH DETECTORS FAIL ON SPANISH

DETECTOR	ASVSPOOF (ENGLISH)	HISPASPOOF (SPANISH)
LFCC+GMM	3.59% EER	42.71% EER
Wav2Vec2-AASIST	0.27% EER	19.92% EER
PaSST	4.77% EER	32.14% EER

Language gap: Best English detector (0.27% EER) jumps to 19.92% on Spanish. This validates our entire thesis premise — language-specific anti-spoofing research is essential.

SPANISH-TRAINED DETECTORS

DETECTOR	HISPASPOOF EER
Spec-ResNet	0.72%
LFCC+GMM	1.57%
PaSST	4.10%
Wav2Vec2-AASIST	10.27%

HABLA 2.0 VS HISPASPOOF

ASPECT	HABLA 2.0	HISPASPOOF
Accents	7 LatAm	6 (incl. Peninsular)
Speakers	1,567	24 + 6 unseen
TTS systems	5 (+ partial spoof)	6
TTS overlap	FishSpeech in both	
Novel TTS	Qwen3-TTS, OuteTTS	ElevenLabs, F5-TTS
Partial spoof	Yes (W1/W2/W3)	No
Quality metrics	WER/CER/NISQA/SIM	EER only
Focus	Voice cloning attacks	Detection + attribution

Key differentiators. (1) 65x more speakers (1,567 vs 24). (2) Qwen3-TTS and OuteTTS are absent from HISPASpoof. (3) Partial spoof is entirely novel for Spanish. (4) We include Venezuelan and Puerto Rican accents; they include Peninsular. (5) Our per-sample quality validation enables fine-grained attack analysis beyond EER.

Partial spoof investigation

Existing anti-spoofing research treats utterances as fully bonafide or fully spoofed. But what if only **1-3 words** are synthetic? This is the partial spoof problem — and no dataset exists for Spanish.

THE GAP

PartialSpoof (Zhang et al., 2023)

English-only (VCTK), segment-level splicing from ASVspoof 2019 LA systems. The only existing partial spoof dataset — but limited to English.

No partial spoof for Spanish

Neither HISPASpoof, LRLSpoof, nor SpeechFake-MD address partially spoofed utterances. This is a completely unexplored area for any variety of Spanish.

OUR APPROACH

Word-level splicing from full clones

Clone the full sentence with a TTS system, then extract individual words via forced alignment and splice them back into the bonafide audio. Preserves natural prosody.

Three tiers: W1 / W2 / W3

Replace 1, 2, or 3 words per utterance. Each tier has a minimum sentence length to keep spoof ratio below 25%. Non-adjacent constraint prevents contiguous fake blocks.

6 attack backends

Same TTS systems used for full-synthesis attacks (FishGram, Qwen, Chatterbox, OpenVoice, OuteTTS, CosyVoice) — enabling comparison of partial vs full spoof per system.

Word-level partial spoof for Spanish

First partial spoof dataset for Latin American Spanish. Unlike full-synthesis attacks, partial spoof replaces **1-3 individual words** in bonafide audio with voice-cloned versions — making detection significantly harder. Inspired by Zhang et al. (IEEE/ACM TASLP 2023) but adapted for Spanish with modern TTS systems.

WORD REPLACEMENT TIERS

TIER	WORDS REPLACED	MIN SENTENCE	MAX SPOOF RATIO
W1	1 word	4 words	25%
W2	2 words	8 words	25%
W3	3 words	12 words	25%

EXAMPLE — W2 SPLICE

Los **presidentes** estan muy peleados por el **ultimo** caso de corrupcion.

Highlighted words extracted from TTS clone and spliced into bonafide audio with 30-80ms crossfade (7 techniques)

Why full-sentence cloning first? Words generated in isolation have flat, citation-form prosody. By cloning the full sentence, then extracting target words via forced alignment, the spliced words carry natural in-context prosody — making the partial spoof harder to detect and scientifically more valuable.

7-STEP PIPELINE

- 01

Transcribe bonafide
Parakeet TDT 0.6b-v3 with word-level timestamps
- 02

Clone full sentence
6 attack strategies (Factory pattern): FishGram, Qwen, Chatterbox, OpenVoice, OuteTTS, CosyVoice
- 03

Forced alignment
Word timestamps for both bonafide and cloned versions
- 04

Select words
Random non-adjacent selection per tier (seeded RNG)
- 05

Splice audio
7 splice techniques, 30-80ms overlap, per-splice random selection
- 06

Validate quality
Parakeet WER/CER + NISQA MOS + ECAPA-TDNN speaker similarity
- 07

Format to ASVspoof2019 LA
FLAC output with **partial_spoof** label in protocol

vs PartialSpoof (Zhang et al., 2023): English-only (VCTK), segment-level splicing from ASVspoof 2019 LA systems. We use word-level extraction from modern TTS (2024-2026), across 7 Latin American accents with 1,567 speakers. First partial spoof dataset for any variety of Spanish.

STATUS

Testing

Qwen strategy validated

TIERS

W1 / W2 / W3

1, 2, or 3 words replaced

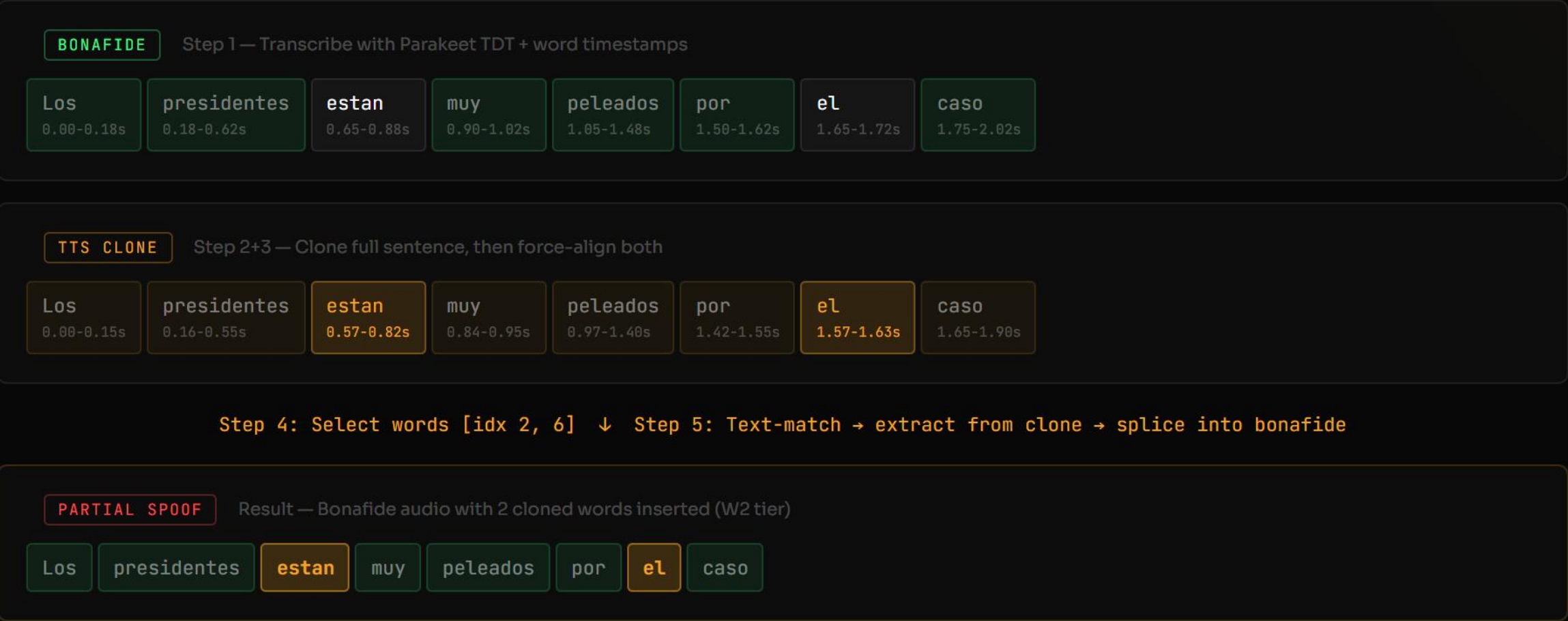
BACKENDS

6

Attack strategies

How word-level splicing works

Clone the full sentence with a TTS system, align both signals at word level, then surgically replace selected words in the bonafide audio with their cloned counterparts.



Key insight: By cloning the *full sentence* first, the extracted words carry natural in-context prosody — not the flat, citation-form prosody you get from generating words in isolation. The splice is harder to detect because the cloned word already "fits" the sentence melodically.

The splice boundary challenge

Replacing a word is not as simple as cutting and pasting waveform samples. The junction between bonafide and cloned audio creates measurable artifacts that both humans and detectors can identify.

WHAT GOES WRONG AT THE BOUNDARY

AMPLITUDE Energy discontinuity

The bonafide and cloned segments have different RMS levels. A sudden jump in loudness at the splice point creates an audible "seam."

PHASE Click artifacts

If the waveform is cut at non-zero amplitude, the instantaneous jump produces a broadband click. Solution: snap cuts to zero-crossings.

SPECTRAL Formant mismatch

The TTS model produces slightly different spectral envelopes than the original speaker. The formant trajectory is discontinuous at the join.

PROSODY FO discontinuity

Fundamental frequency (pitch) trajectories differ between bonafide and clone. An abrupt FO jump is perceptually unnatural, even if amplitude is matched.

OUR MITIGATION STRATEGY

Zero-crossing alignment

Snap all 4 splice boundaries (bonafide start/end, cloned start/end) to the nearest zero-crossing within +/-5ms. Eliminates click artifacts.

RMS energy normalization

Scale the cloned segment's amplitude to match the bonafide region's RMS level. Removes the loudness seam.

7 crossfade techniques

Randomly select a fade curve and overlap duration (30-80ms) per splice. Prevents detectors from learning a fixed crossfade fingerprint.

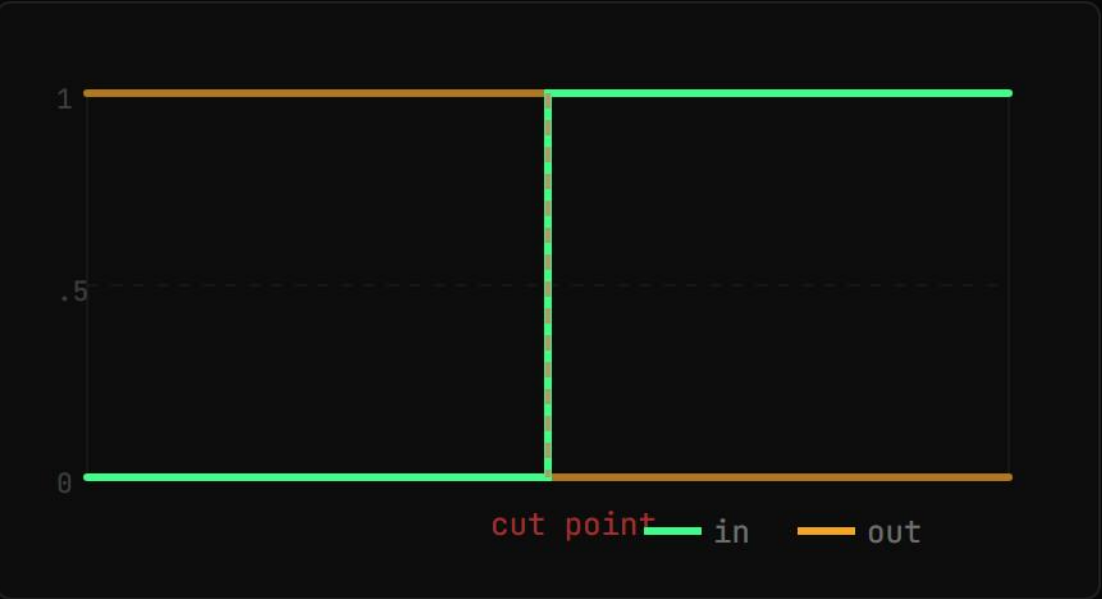
Gap-limited margins

Crossfade overlap is clamped to the available silence between adjacent words. The cloned word content is *never* truncated by the fade.

Key finding (Negroni et al., 2024): Even with optimal crossfading, purely spectral analysis achieves 6.16% EER on Spoof with zero trained parameters. Artifacts persist — the goal is diversity, not invisibility.

Direct cut-paste

Hard concatenation at zero-crossings with no overlap or fading. The simplest and most forensically detectable method.



NO OVERLAP

10% OF SPLICES

METHOD

Abrupt concatenation at the splice point. Overlap = 0 ms. No fade curves applied. Bonafide segment ends, cloned segment begins immediately.

SOURCE

PartialSpoof (Zhang et al., 2023), HAD (Yi et al., 2021). Both use simple concatenation as their baseline splicing method.

ACOUSTIC EFFECT

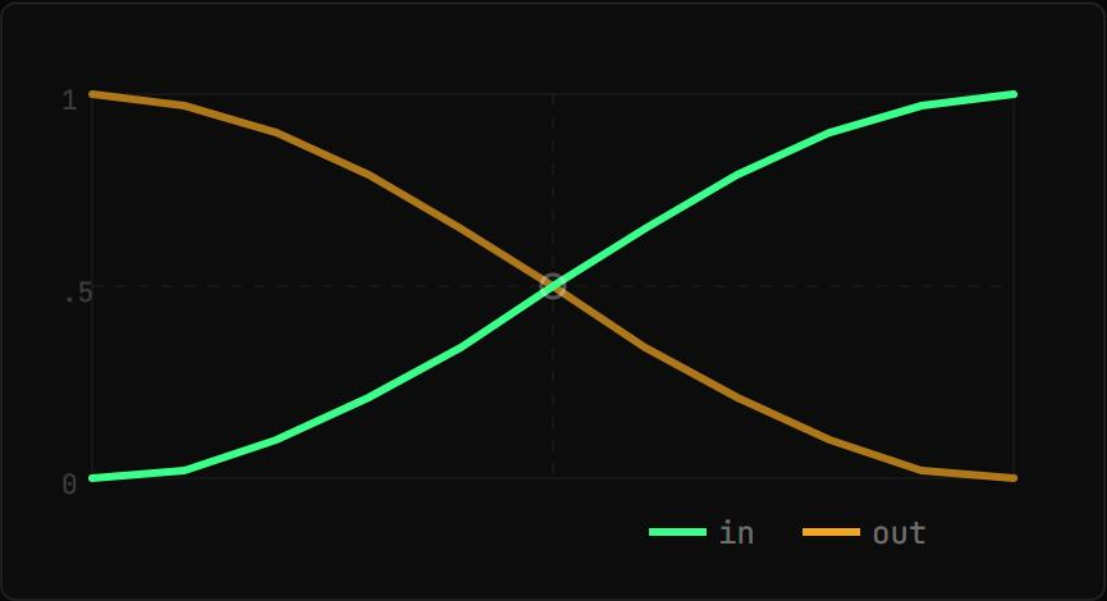
Produces energy discontinuities and potential phase jumps at the boundary. Even with zero-crossing alignment, spectral differences between segments create a measurable forensic trace. **Most detectable technique** — Negroni et al. achieves 6.16% EER with zero training.

WHY INCLUDE IT?

Forces detectors to handle the easiest case. If a model cannot detect cut-paste splices, it has no chance with smoother methods. Serves as the **forensic floor**.

OLA Hanning

Overlap-Add with a Hann window. The gold-standard in signal processing for smooth segment transitions. Equal-gain S-curve blend.



EQUAL-GAIN

20% OF SPLICES

FORMULA

$$\text{fade_in}(t) = 0.5 * (1 - \cos(\pi * t))$$

$$\text{fade_out}(t) = 0.5 * (1 + \cos(\pi * t))$$

SOURCE

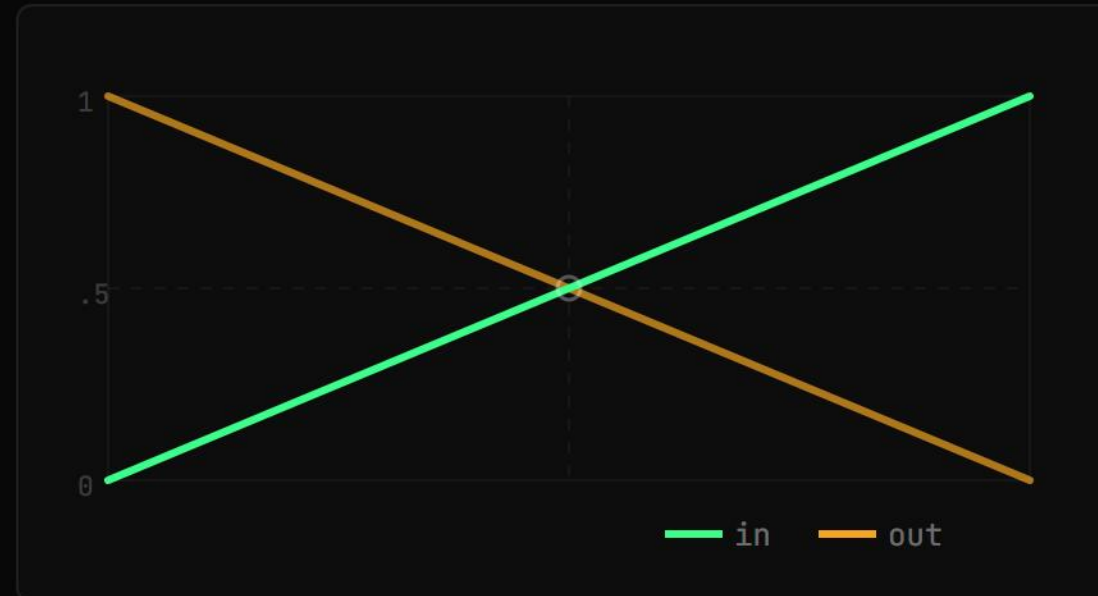
Negroni et al. (arXiv:2408.13784, 2024) — tested OLA with Hann windows from 256 to 4096 samples. Found minimum 1024 samples (64ms) needed to effectively mask artifacts. Even at 4096 samples, AUC remains 88.99%.

ACOUSTIC EFFECT

S-shaped blend: slow transition at start and end, fast in the middle. Perceptually the smoothest technique. **fade_in + fade_out = 1.0** everywhere (constant total gain), so no amplitude dip or boost in the overlap.

Linear crossfade

The simplest crossfade: a straight diagonal blend. Equal-gain but with a perceptual loudness dip at the midpoint.



EQUAL-GAIN

15% OF SPLICES

FORMULA

$$\text{fade_in}(t) = t$$

$$\text{fade_out}(t) = 1 - t$$

SOURCE

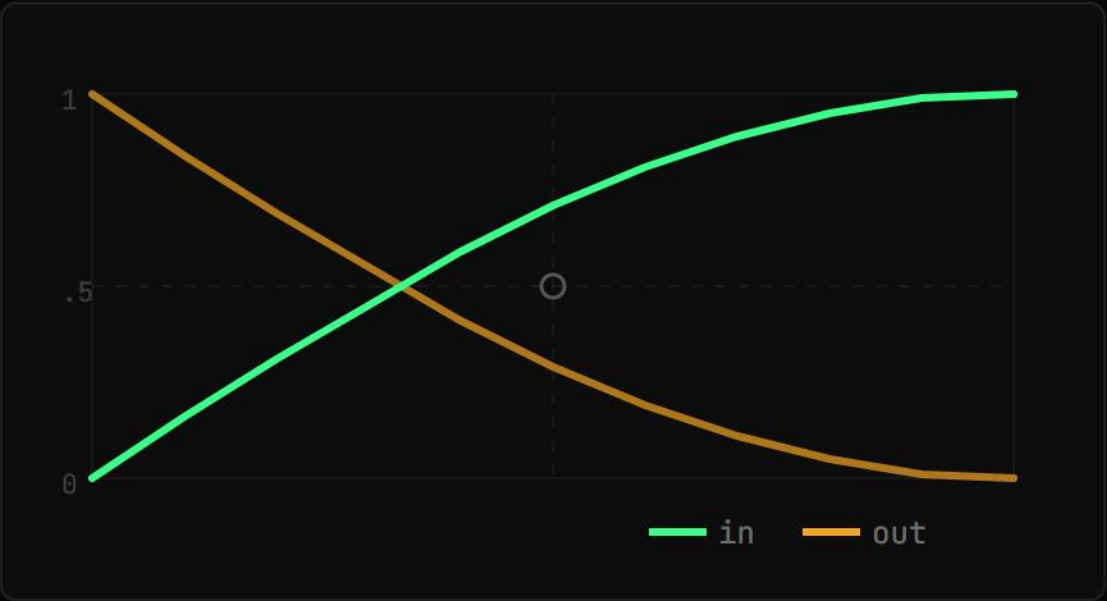
LlamaPartialSpoof (Luong et al., ICASSP 2025) — one of 5 crossfade functions randomly assigned per splice. Also the baseline in Negroni et al.'s "triangular window" analysis.

ACOUSTIC EFFECT

Constant total amplitude ($\text{in} + \text{out} = 1.0$), but perceived loudness dips at $t=0.5$ because **power** = $0.5 + 0.5 = 0.5$ (half of original). This "loudness hole" is the main perceptual artifact and the main forensic signature of linear crossfades.

Cosine (equal-power)

Quarter-cosine crossfade with constant total power. The broadcast-standard crossfade: no loudness dip, no loudness boost.



EQUAL-POWER

20% OF SPLICES

FORMULA

$$\text{fade_in}(t) = \sin(\pi * t / 2)$$

$$\text{fade_out}(t) = \cos(\pi * t / 2)$$

SOURCE

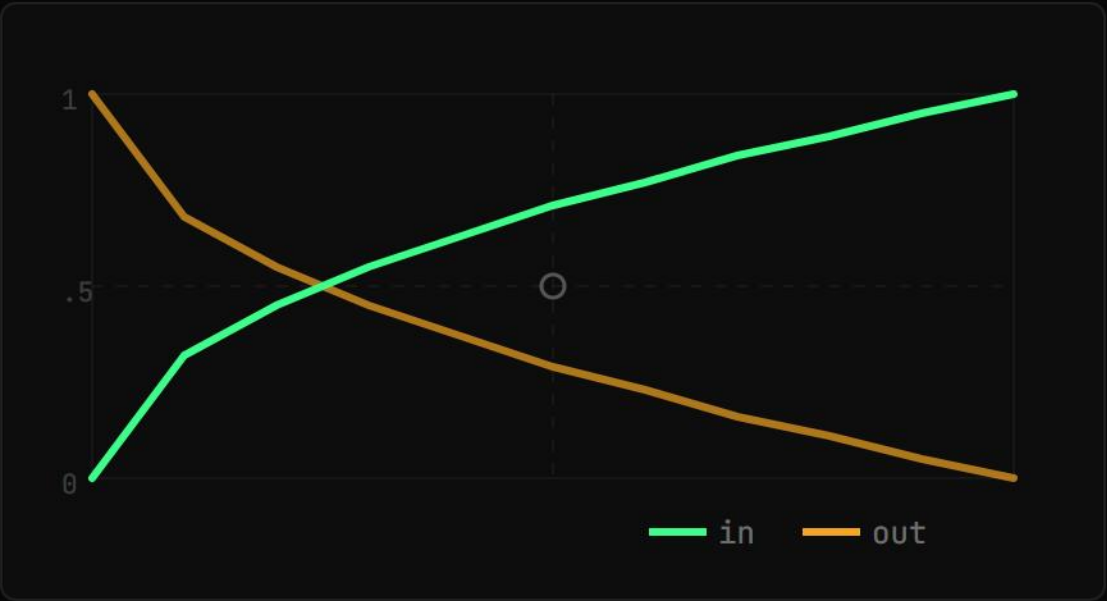
LlamaPartialSpoof — "quarter sine" variant. Also HQ-MPSD (Li et al., 2025) uses a cosine window with 30ms overlap as their default technique.

ACOUSTIC EFFECT

Constant total power: **$\sin^2 + \cos^2 = 1$** . No perceptual loudness dip (unlike linear). The transition starts gradually, accelerates, then the outgoing signal drops quickly. Harder to detect than Hann because the total amplitude sum exceeds 1.0 (slight boost at midpoint), masking energy discontinuities.

Square-root (half-sine)

Equal-power crossfade via square-root law. Aggressively shifts energy to the incoming signal early in the transition.



EQUAL-POWER

15% OF SPLICES

FORMULA

$\text{fade_in}(t) = \sqrt{t}$
 $\text{fade_out}(t) = \sqrt{1 - t}$

SOURCE

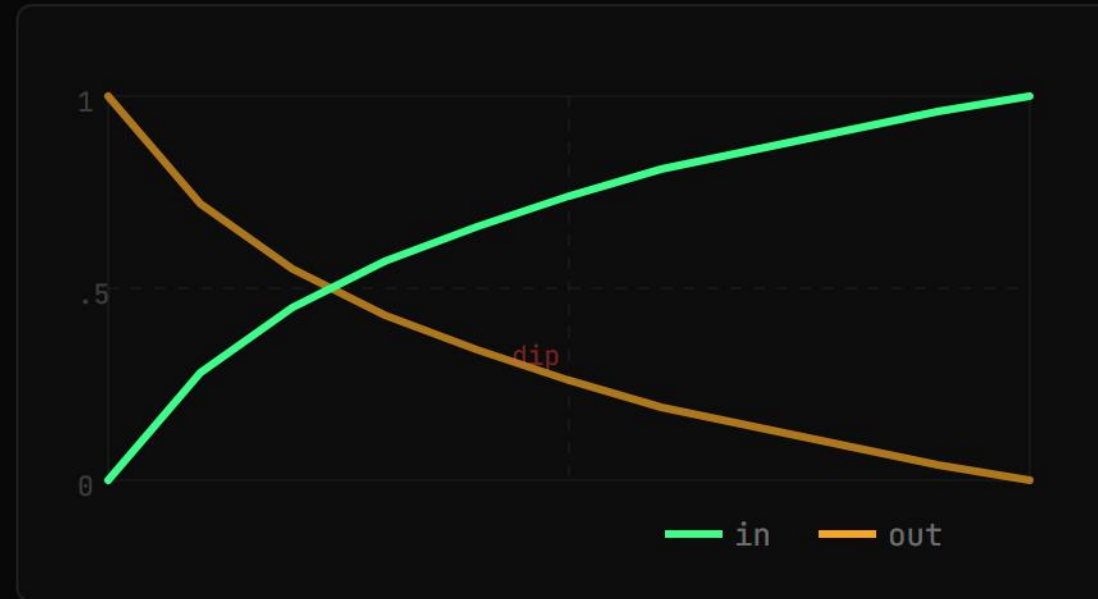
LlamaPartialSpooF — "half sine" variant. Common in broadcast/DAW production as the "constant power" crossfade. We implement it as the algebraic square-root law for numerical stability.

ACOUSTIC EFFECT

Very fast initial rise: at $t=0.1$, the incoming signal is already at 31.6% amplitude (vs 15.6% for cosine). The outgoing signal drops rapidly too. Creates a **"snap" to the new signal** early in the overlap, leaving most of the crossfade as fine adjustment. Equal power: $t + (1-t) = 1$.

Logarithmic crossfade

Aggressive initial rise then plateau. Neither equal-gain nor equal-power — creates a unique energy dip artifact.



ENERGY DIP

10% OF SPLICES

FORMULA

$$\text{fade_in}(t) = \log(1 + 9t) / \log(10)$$

$$\text{fade_out}(t) = \log(1 + 9(1-t)) / \log(10)$$

SOURCE

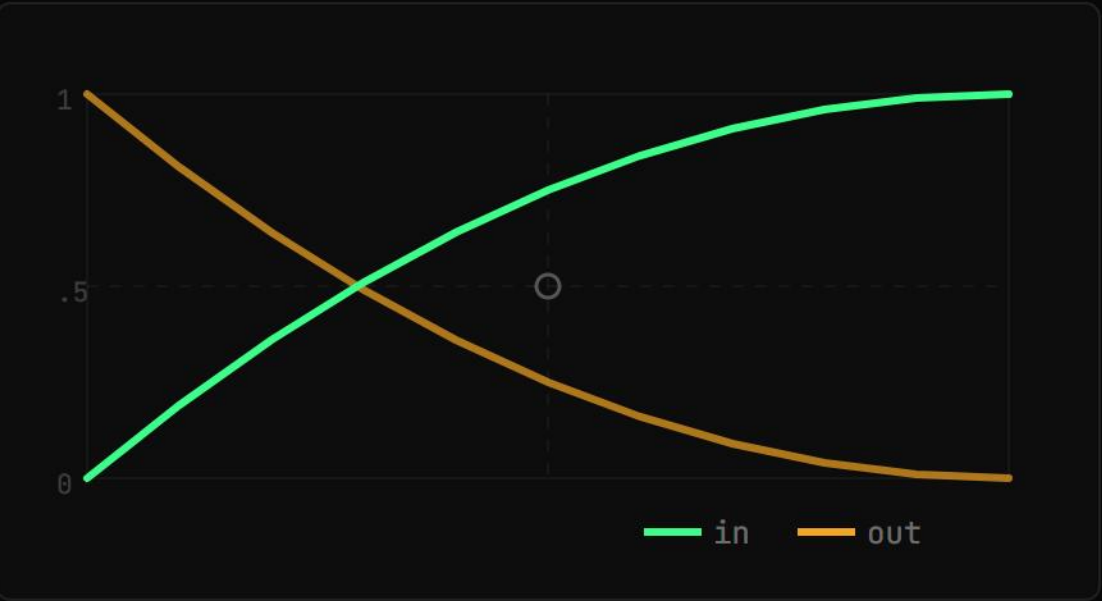
LlamaPartialSpoof — "logarithmic" variant. One of the 5 fading functions tested in their crossfade-vs-EER ablation (Table V).

ACOUSTIC EFFECT

Neither equal-gain nor equal-power. At $t=0.5$: $\text{in} + \text{out} = 0.740 + 0.740 = 1.48$ (amplitude), but $\text{in}^2 + \text{out}^2 = 0.548 + 0.548 = 1.096$ (near-equal power). The energy dip is the most distinctive forensic fingerprint among the 7 techniques — detectors can learn this as a characteristic "log signature."

Inverted parabola

Concave quadratic blend. Equal-gain with a smooth acceleration profile, sitting between linear and Hann in perceptual smoothness.



EQUAL-GAIN

10% OF SPLICES

FORMULA

$$\text{fade_in}(t) = 1 - (1 - t)^2 = 2t - t^2$$

$$\text{fade_out}(t) = (1 - t)^2$$

SOURCE

LlamaPartialSpoof — "inverted parabola" variant. The quadratic complement to their logarithmic and linear fades. Named for the shape of $1 - (1 - t)^2$ which is a downward-opening parabola.

ACOUSTIC EFFECT

Equal-gain: $(2t - t^2) + (1 - t)^2 = 1.0$. Rises faster than linear at the start (concave shape) but slower than logarithmic. Perceptually intermediate: smoother than linear but without the S-curve smoothness of Hann. The fast initial blend makes it more like "swapping" the signal with a gentle fade.

Technique distribution & literature gaps

Seven techniques with varied proportions produce a forensically diverse dataset. Each splice independently draws its method and overlap duration, preventing detectors from learning a single crossfade fingerprint.

PER-SPLICE DISTRIBUTION

TECHNIQUE	PROPORTION	ENERGY	OVERLAP
Direct cut-paste	10%	NONE	0 ms
OLA Hanning	20%	EQUAL-GAIN	30-80 ms
Linear	15%	EQUAL-GAIN	30-80 ms
Cosine	20%	EQUAL-POWER	30-80 ms
Square-root	15%	EQUAL-POWER	30-80 ms
Logarithmic	10%	DIP	30-80 ms
Inv. parabola	10%	EQUAL-GAIN	30-80 ms

Design rationale: Equal proportions for the two SOTA methods (OLA Hanning 20%, Cosine 20%). Lower proportions for extreme cases (cut-paste 10%, log 10%, parabola 10%). LlamaPartialSpoof assigns 30-80ms uniformly — we follow this anchor.

UNSOLVED PROBLEMS (THESIS CONTRIBUTIONS)

1. Zero-gap boundary policy

No pipeline specifies behavior when adjacent cloned words leave <30ms of bonafide margin. Five strategies exist; none is attested in any paper.

2. F0 discontinuity

No partial-spoof paper reports pitch discontinuity at splice boundaries. Publishing F0 histograms at MFA word boundaries would be novel.

3. Crossfade duration vs EER

LlamaPartialSpoof's 30-80ms range is the only anchor. No paper maps overlap length to detection performance — a tractable publishable experiment.

4. No source code released

None of the 4 canonical pipelines (PartialSpoof, LlamaPartialSpoof, HAD, HQ-MPSD) releases their splicer code. Our pipeline is fully reproducible.

OUR PIPELINE VS EXISTING PARTIAL SPOOF DATASETS

FEATURE	PARTIALSPOOF	LLAMAPARTIALSPOOF	HAD	HQ-MPSD	OURS (HABLA 2.0)
Language	English	English	Chinese	6 languages	Spanish (7 accents)
Splice level	Segment (VAD)	Word (MFA)	Character	Word (MFA)	Word (Parakeet TDT)
Crossfade types	1 (simple OLA)	5 shapes	2 (cosine, cut)	1 (cosine 30ms)	7 techniques
Overlap range	VAD margins	30-80ms uniform	None	Fixed 30ms	30-80ms uniform
TTS systems	ASVspoof 2019 LA	XTTS, Parler, Llama	7 legacy	3 modern	6 modern (2024-2026)
Speakers	107 (VCTK)	~40 (VCTK)	281	4 / language	1,567 (HABLA v2)
Code released	No	No	No	No	Yes (open source)

The real problem: alignment, not crossfade

After implementing all 7 crossfade techniques and listening to the results, we discovered that **all techniques sounded identical**. The audible problem was not the fade curve — it was three deeper issues.

1. Blind word selection

Step 4 picked words **randomly** without checking if they could be cleanly cut from the clone. Words at fluid speech boundaries (no pause between them) sounded terrible when spliced.

2. No clone quality gate

Bad TTS clones (low speaker similarity) passed through the entire pipeline, producing obviously detectable splices regardless of crossfade quality.

3. Duration mismatch

A 480ms cloned word replacing a 640ms bonafide word shifted everything after by 160ms. Speech rhythm was completely destroyed.

ENERGY AT "DAME" → "UNICAMENTE" BOUNDARY (0.80S) — CLONED AUDIO



No energy valley. The TTS generates "Dame unicamente" as continuous speech — there is no pause to cut at. The aligner places the boundary at 0.80s, but the energy never drops below 0.097. The "u" of "unicamente" ends up inside "Dame".

Key insight: The crossfade technique (Hann, cosine, linear, etc.) only smooths the last 15-60ms of the junction. But if the *word itself* is cut wrong or doesn't fit the time slot, no crossfade can fix it. The problem is upstream of the crossfade.

Valley-score selection + clone similarity gate

Three fixes that address the root causes: select only words with clean boundaries, reject bad clones early, and preserve speech rhythm by matching durations exactly.

FIX 1: VALLEY-SCORE WORD SELECTION

FORMULA

$$\text{score} = \text{min_rms} / \text{avg_rms}$$

For each word boundary: compute RMS energy in 5ms frames within ±100ms. The ratio of the minimum frame to the average tells us how deep the valley is. Score 0.0 = perfect silence. Score 1.0 = no valley at all.

RANKING: "DAME UNICAMENTE EL DATO MAS RELEVANTE"

#1	"el"		0.165
#2	"relevante"		0.179
#3	"dato"		0.210
#5	"Dame"		0.700
#6	"unicamente"		0.700

Words above threshold (0.65) are never selected. "Dame" and "unicamente" are rejected because their shared boundary has no energy valley.

FIX 2: CLONE SIMILARITY GATE

Before alignment and splicing, compute **ECAPA-TDNN cosine similarity** between the bonafide reference and the TTS clone. Clones with SIM < 0.60 are rejected immediately — saving compute and preventing obviously bad splices.

THRESHOLD

0.60

EMBEDDING

192-d

FIX 3: DURATION-PRESERVING SPLICE

The cloned word is **time-stretched to fit the exact bonafide slot**. Total audio length never changes. Gaps between words are preserved. Speech rhythm stays intact.

BEFORE VS AFTER

Before: cut 640ms, insert 480ms → -160ms shift
After: stretch 480ms → 640ms → 0ms shift

EDGE CASES HANDLED

All bad scores

Reject sample entirely

Word < 200ms

Ineligible (too short)

Stretch > 1.25x

Skip word (distortion)

First/last word

Only internal boundary scored

← prev

next →

Two-gate pre-filter before splicing

Before any audio is cut and pasted, two independent filters ensure only high-fidelity, speaker-faithful clones of words with clean acoustic boundaries are accepted.



GATE 1 — Clone Similarity

METHOD

ECAPA-TDNN (192-d embedding) cosine similarity between the bonafide reference speaker and the generated TTS clone. Applied **before alignment and splicing** to avoid wasting compute on bad clones.

THRESHOLD & REJECTION RATES

FishGram	Qwen3-TTS	OpenVoice
0.602	0.720	0.394
avg SIM	avg SIM	avg SIM

```
sim = cosine(emb_bonafide, emb_clone)
if sim < 0.60: reject
```

GATE 2 — Valley Score Word Selection

METHOD

For each word boundary, measure how deep the energy valley is. Only words at **naturally quiet boundaries** are eligible for replacement. This prevents splicing at fluid speech points where no natural cut exists.

FORMULA

$score = \frac{min_rms}{avg_rms}$
Computed in 5ms frames within $\pm 100ms$ of the boundary. **0.0** = perfect silence. **1.0** = no valley. **Threshold: 0.65** — words above are ineligible.

ADDITIONAL ELIGIBILITY CONSTRAINTS

duration $\geq 200ms$ — no short function words
stretch $\in [0.80, 1.25]$ — time-stretch within $\pm 25\%$
non-adjacent — selected words ≥ 2 apart

Homogenize boundary anomalies so the splice no longer stands out

After Step 5 splices the cloned word, the only unusual boundaries are the two around the spoof. A detector exploits "find the noisy boundary" as a shortcut. Step 5b applies the same kind of structural manipulations to **every** internal boundary so the spoof no longer stands out.

Negroni et al. (2024): 6.16 % EER on PartialSpoof with zero training, purely by analyzing dynamic-range discontinuity at the splice join.
Muller (2024): detectors generalize via dataset-specific shortcuts (silence length, spectral cleanness), not synthesis features.

PER-BOUNDARY INDEPENDENT COIN FLIP

```
for each internal boundary i (right-to-left):
  if rng.uniform() < JITTER_PROBABILITY = 0.5:
    op = uniform_pick(truncate, overlap, bleed)
    apply(audio, op, uniform_magnitude_within_range)
  else: leave boundary natural
```

Spoof boundaries receive the **same coin flip** on top of the splice. Seed = JITTER_SEED + sha256(splice_key)[:4] for reproducibility.

WHY P = 0.5: COUNT-DISTRIBUTION MIXING

BEFORE JITTER (DETERMINISTIC)		AFTER JITTER (P = 0.5)	
BOUNDARY TYPE	MANIPULATIONS	BOUNDARY TYPE	DISTRIBUTION OVER COUNT
bonafide-bonafide	0 (always)	bonafide-bonafide	0 (50 %) or 1 (50 %)
spoof	1 (always)	spoof	1 (50 %) or 2 (50 %)
Detector classifies trivially on count.		Both share equal mass at count = 1 → ambiguous.	

Word interiors stay intact. All three operations act only at boundaries -- never inside a word. With p = 0.5 per boundary, the probability that both flanking boundaries of a word flip natural is 0.25. By construction, **~25 % of non-spoof words remain bit-identical to the bonafide**. Intelligibility is preserved; only the boundary fingerprint is homogenized.

Truncate, overlap, bleed: chosen uniformly when the coin flips heads

Each operation imitates a specific splice artifact, so a manipulated bonafide-bonafide boundary is acoustically indistinguishable from a real splice boundary at the spectral level. Magnitudes are drawn uniformly at random within ranges anchored to Spanish phonetics and to the LlamaPartialSpoof crossfade convention.

OPERATION	MAGNITUDE	EFFECT ON AUDIO	MIMICS SPLICE ARTIFACT
truncate	10 - 40 ms	Cut left tail or right head → abrupt onset/offset	Hard cut/paste splice (no crossfade)
overlap	30 - 80 ms	Sum left tail with right head, Hanning fade	OLA-Hanning crossfade splice
bleed	20 - 60 ms	Insert fragment of one word into the other → pre/post-echo	Tail bleed of crossfade splice

PER-BOUNDARY PROBABILITY TABLE

```
P(natural) = 0.500
P(truncate left_tail) = 0.083
P(truncate right_head) = 0.083
P(overlap hanning) = 0.167
P(bleed L→R) = 0.083
P(bleed R→L) = 0.083
-----
total = 1.000
```

Uniform pick among ops because we have no prior knowledge of which artifact type the detector is most sensitive to — that would require running the detector first (circular).

MAGNITUDE GROUNDING

- Overlap (30-80 ms) matches the LlamaPartialSpoof crossfade range (Luong et al. 2024) exactly. Same temporal scale as the splice itself.
- Truncate (10-40 ms) stays below Spanish syllable nucleus duration (~50-90 ms) so intelligibility is preserved. 10 ms minimum is at the threshold of audibility.
- Bleed (20-60 ms) covers Spanish VOT for voiceless stops (4-29 ms) plus consonant-vowel transition — inserted fragment sounds like a partial phoneme, not a full second phoneme.

"Necesito que me des informacion sobre el mural..."

Speaker arf_00295, 18 words, 1 cloned word ("Necesito"). 17 internal boundaries. After Step 5b: 2 truncate, 3 overlap, 4 bleed, 8 natural. Drift -46 ms.

BONAFIDE (ORIGINAL)

▶ 0:00 / 0:09

🔊 ⋮

SPLICED W1 (NO JITTER)

▶ 0:00 / 0:09

🔊 ⋮

JITTERED W1J

▶ 0:00 / 0:09

🔊 ⋮

#	WORD	SPAN (S)	LEFT BOUNDARY	RIGHT BOUNDARY	STATUS
1	Necesito	1.20 - 1.84	utterance start	overlap 51.5 ms	SPOOF
2	que	1.84 - 2.08	overlap 51.5 ms	natural	touched L
3	me	2.08 - 2.24	natural	bleed 42.2 ms L→R	touched R
4	des	2.24 - 2.56	bleed 42.2 ms	overlap 50.1 ms	touched both
5	informacion	2.56 - 2.88	overlap 50.1 ms	bleed 45.5 ms L→R	touched both
6	sobre	2.88 - 3.12	bleed 45.5 ms	truncate 16.9 ms RH	touched both
7	el	3.12 - 3.28	truncate 16.9 ms RH	bleed 28.1 ms L→R	touched both
8	mural	3.28 - 3.60	bleed 28.1 ms	natural	touched L
9	que	3.60 - 3.76	natural	natural	CLEAN
10	hicieron	3.76 - 4.08	natural	natural	CLEAN
11	Frida	4.08 - 4.48	natural	natural	CLEAN
12	Kahlo	4.48 - 4.88	natural	truncate 25.0 ms RH	touched R
13	y	4.88 - 4.96	truncate 25.0 ms RH	natural	touched L
14	Diego	4.96 - 5.28	natural	bleed 24.8 ms R→L	touched R
15	Rivera	5.28 - 5.68	bleed 24.8 ms	overlap 42.9 ms	touched both
16	antes	5.84 - 6.08	overlap 42.9 ms	natural	touched L
17	de	6.08 - 6.24	natural	natural	CLEAN
18	morir	6.24 - 6.80	natural	utterance end	CLEAN

Tally. 1 spoof, 5 clean (29 %), 12 touched. Note **boundary 0 (1.84 s)**: it is BOTH the right edge of "Necesito" (the spoof crossfade) AND received an overlap jitter on top -- exactly the boundary a detector would target as "the splice", now stacked with a second manipulation. The 3-word clean cluster "**que hicieron Frida**" (3.60-4.48 s) is a stretch where the audio is pure bonafide passthrough.

"Segun la television y los medios, hubieron bastantes muertos por el terremoto."

Speaker arf_00295, 12 words, 1 cloned word ("muertos"). 11 internal boundaries. After Step 5b: 3 truncate, 0 overlap, 2 bleed, 6 natural. Drift +24 ms.

BONAFIDE (ORIGINAL)

▶ 0:00 / 0:06

🔊 ⋮

SPLICED W1 (NO JITTER)

▶ 0:00 / 0:06

🔊 ⋮

JITTERED W1J

▶ 0:00 / 0:07

🔊 ⋮

#	WORD	SPAN (S)	LEFT BOUNDARY	RIGHT BOUNDARY	STATUS
1	Segun	0.32 - 1.04	utterance start	bleed 44.9 ms R→L	touched R
2	la	1.04 - 1.20	bleed 44.9 ms	bleed 31.4 ms R→L	touched both
3	television	1.20 - 1.76	bleed 31.4 ms	natural	touched L
4	y	1.76 - 1.92	natural	natural	CLEAN
5	los	1.92 - 2.00	natural	truncate 13.2 ms RH	touched R
6	medios,	2.00 - 2.32	truncate 13.2 ms RH	natural	touched L
7	hubieron	2.48 - 2.96	natural	natural	CLEAN
8	bastantes	2.96 - 3.36	natural	natural (= spoof start)	CLEAN
9	muertos	3.36 - 3.68	natural (jitter tails)	truncate 10.1 ms RH	SPOOF
10	por	3.68 - 3.84	truncate 10.1 ms RH	natural	touched L
11	el	3.84 - 4.00	natural	truncate 28.8 ms RH	touched R
12	terremoto.	4.00 - 4.88	truncate 28.8 ms RH	utterance end	touched L

Tally. 1 spoof, 3 clean (27 %), 8 touched. **Spoof boundary at 3.36 s flipped natural** -- the cloned "muertos" enters via crossfade splice but the jitter coin landed tails on its left edge. Adjacent word "bastantes" therefore qualifies as CLEAN despite being right next to the spoof. The right edge of "muertos" (3.68 s) did get a 10 ms truncate stacked on top of bleed from previous word, but no overlaps -- pure truncate-and-bleed sample for this random plan.

Same source as Audio 2, two cloned words ("hubieron" + "muertos")

Speaker arf_00295, 12 words, 2 cloned. 11 internal boundaries. Different RNG seed than W1 (seeded by splice_key) so a different jitter plan emerges. After Step 5b: 3 truncate, 1 overlap, 1 bleed, 6 natural. Drift -67 ms.

BONAFIDE (ORIGINAL)

▶ 0:00 / 0:06

🔊 ⋮

SPLICED W2 (NO JITTER)

▶ 0:00 / 0:06

🔊 ⋮

JITTERED W2J

▶ 0:00 / 0:06

🔊 ⋮

#	WORD	SPAN (S)	LEFT BOUNDARY	RIGHT BOUNDARY	STATUS
1	Segun	0.32 - 1.04	utterance start	natural	CLEAN
2	la	1.04 - 1.20	natural	natural	CLEAN
3	television	1.20 - 1.76	natural	natural	CLEAN
4	y	1.76 - 1.92	natural	natural	CLEAN
5	los	1.92 - 2.00	natural	overlap 39.6 ms	touched R
6	medios,	2.00 - 2.32	overlap 39.6 ms	bleed 43.1 ms R→L (= spoof start)	touched both
7	hubieron	2.48 - 2.96	bleed 43.1 ms (splice + jitter)	truncate 29.6 ms RH (splice + jitter)	SPOOF
8	bastantes	2.96 - 3.36	truncate 29.6 ms RH (= spoof end of hubieron)	truncate 30.9 ms RH (= spoof start of muertos)	SANDWICHED
9	muertos	3.36 - 3.68	truncate 30.9 ms RH (splice + jitter, cut_paste)	natural (jitter tails, cut_paste)	SPOOF
10	por	3.68 - 3.84	natural (= spoof end of muertos)	natural	CLEAN
11	el	3.84 - 4.00	natural	truncate 10.3 ms RH	touched R
12	terremoto.	4.00 - 4.88	truncate 10.3 ms RH	utterance end	touched L

Tally. 2 spoof, 5 clean (50 %), 4 touched, 1 sandwiched. The 4-word clean run "**Segun la television y**" at the start (0.32-1.92 s) is pure bonafide passthrough. "**bastantes**" (between the two spoofs) takes hits on both sides because both spoof boundaries flipped heads -- it ends up with two truncates flanking it. The cloned "muertos" was spliced with cut_paste (the rare 10 % no-crossfade fade), and its left boundary stacks the cut_paste with a 30.9 ms truncate jitter -- the most aggressive single boundary in the three samples. Listen carefully there.

Bonafide vs. Partially Spoofed Speech

Speaker arf_00295 (Argentina Female). Same utterance. Words replaced with voice-cloned versions using valley-score selection and duration-preserving splice.

ORIGINAL · BONAFIDE

arf_00295

Argentina · Female · HABLA v2

"Hay varios cines cerca del hotel. ¿Estás buscando algo en especial?"

▶ 0:00 / 0:06

—

🔊

⋮

Spoofed words

SpooF ratio

none

0.0%

PSW1 · PARTIAL SPOOF

W1

1 word replaced

Qwen3-TTS · linear crossfade · 58.8ms

"Hay varios cines **cerca** del hotel. ¿Estás buscando algo en especial?"

▶ 0:00 / 0:06

—

🔊

⋮

Valley score

Stretch ratio

WER · NISQA · SIM

0.142

1.00x

0% · 4.81 · 0.992

PSW2 · PARTIAL SPOOF

W2

2 words replaced

Qwen3-TTS · logarithmic + half_sine

"Hay **varios** cines **cerca** del hotel. ¿Estás buscando algo en especial?"

▶ 0:00 / 0:06

—

🔊

⋮

Valley scores

Stretch ratios

WER · NISQA · SIM

0.118 · 0.142

1.00x · 1.00x

0% · 4.81 · 0.961

HOW WORDS ARE SELECTED

VALLEY SCORE RANKING FOR THIS UTTERANCE

#1

"varios"

0.118

#2

"cerca"

0.142

#3

"cines"

0.430

×

"Hay"

0.006

"Hay" rejected: stretch ratio 0.67 outside [0.80, 1.25] — too short in clone.

VALIDATION SUMMARY (7 SAMPLES)

AVG WER

2.27%

vs 30% ceiling

AVG NISQA

4.66

high quality

AVG SIM

0.885

speaker preserved

PASS RATE

7 / 7

validation (arf_00295)

Audio files are relative to project root on ml-server03. Open presentation.html from ~/ANTI-SPOOFING-VOICE-LATIN-AMERICA/ to play audio.

Next steps

01

Attack production runs — 5 of 6 complete

CHATTERBOX 59%

Done: FishGram (95.2% pass), Qwen (87.9%), OpenVoice (83.4%), OuteTTS (72.3%), OmniVoice (93.9%). ~155K passed samples across 1,567 speakers. Chatterbox at 21,192 / 35,927 (59%), ETA early June.

02

Partial spoof multi-backend production

NEXT

Attack mix **50% OmniVoice / 30% Qwen / 20% FishGram** — the three highest-fidelity attacks. Extend per-speaker file partition to 3-way so each backend draws from disjoint bonafide utterances. Same 7-step pipeline (Parakeet -> clone -> MMS align -> valley-score select -> splice -> validate -> format).

03

Boundary jitter datasets at scale

NEXT

Qwen jitter pilot validated end-to-end. Produce full-scale jitter datasets for Qwen and OmniVoice ('_jitter' folders, system_id '*_PSW{N}J', audio_id 16M-18M) using the disjoint "jitter" partition. Two new datasets alongside the main partial_spoof.

04

Augmentation pilot — validate structure, plan percentages

NEXT

Pilot codec (MP3/Opus low bitrate), additive noise (MUSAN), and RIR convolution on a small subset to confirm the final dataset structure (train/dev/eval, protocol files) survives augmentation cleanly. Then plan per-augmentation percentage targets for the production dataset (e.g. 30/30/40 clean/codec/noise).

05

Continue writing the paper

Continue thesis paper. New technical contributions to document: **boundary jitter** as defense against the "find the noisy boundary" detector shortcut (Negroni 2024); **prefix-bleed detector + reference-cut fix** for diffusion-LM TTS (OmniVoice); first partial-spoof dataset for Latin American Spanish with 6-TTS architectural diversity.

← prev

next →